

The Semantics of Replacement

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Introduction

Expressions of replacement are of considerable interest philosophically and linguistically, though they have never received any attention in syntax, semantics, philosophy of language or descriptive metaphysics. Expressions of replacement in English include the verb *replace* and *instead*-phrases.¹

- (1) The board replaced the director of the company.
- (2) a. Instead of a man, the firm hired a woman.
 b. the firm that hired a woman instead of a man
- (3) Instead of people dancing in the street, there was silence everywhere.

Replacement as such can be viewed as a formal operation on structured wholes, removing an entity that occupies a particular position or slots within a structured whole and putting another entity into its place.

In (1a), the structured whole targeted by *replace* is a company, involving the substitution of an individual in a particular role within it. In (2a), the *instead*-phrase targets the structured meaning of a sentence, a structured proposition, involving the substitution of a generalized quantifier by another in a slot aligned with the subject position of the sentence. In (2b), the replacement targets a structured type of object, the type of firm that hires a man, and involves substitution of the generalized quantifier that occurs in it by another. The structured whole here is a type representing objects, a structured whole that, like structured propositions, is obtained compositionally, though not from a sentence but a definite description. In (3), the *instead*-phrase targets a background situation, a context in which people dancing in the street was expected. The replacement then involves exchanging that event by another (there being

¹ In addition, there is the somewhat special construction *in your place*, which this paper will only briefly touch upon.

silence everywhere). The structured whole here is a background situation, which is a semantic object construed through a pragmatic process, which serves the purpose of ‘pragmatic enrichment’ for the evaluation of the entire sentence.

The examples illustrate that replacement can apply to both ordinary objects and semantic objects. The verb *replace* targets structured ‘ordinary objects’, whereas *instead*-phrases target structured semantic objects, namely structured propositions (or states of affairs), structured types of objects, and background situations.

This paper will outline a unified semantics of replacement expressions as expressions that apply to structured wholes, namely ordinary objects, structured propositions, structured types, and situations. At the center of the semantics is that of a position, role, place, or slot, a central feature of replacement.

Expressions of replacement are of particular interest for metaphysics. First, they display the importance of a unified notion of structured wholes, which comprises both structured wholes that are ordinary objects and structured wholes that are semantic objects. Expressions of replacement are also of considerable interest to the debate regarding positionalism in the metaphysics of relations (Fine 2000). Not only do slots play a central role in the semantics of replacement, they are also reflected overtly, in the noun *place*, as in *replace* and *in its place*, as well as *in-stead*, where *stead* derives historically from a root for ‘place’. Finally, expressions of replacement give further support for higher-order metaphysics (Skiba 2021). Expressions of replacement when applying to structured propositions or types can target the entire range of higher-order semantic objects for the purpose of substitution: properties, relations, predicate modifiers, generalized quantifiers, etc. They need to do so while respecting the type of position such objects may occupy in the relevant structures.

In what follows, I will establish a number of semantic and syntactic generalizations about expressions of replacement and sketch formal analyses within a unified formal semantics of replacement constructions applying to structured wholes of various kinds.

1. Verbs of replacement

There are two verbs of replacement in English: *replace* and *exchange*. I will focus on *replace*. The verb *replace* describes the removal of an entity from a particular slot or position within a structured whole and putting another entity into that slot. There are various structured wholes that permit replacement of a part:

An artefact:

(4) a. John replaced the leg (of the table).

A spatial region:

b. John replaced the picture on the wall.

A social institution:

c. John replaced the chairman (of the company).

A planned event (Ardisson et al. 2025):

d. John replaced the priest (for the ceremony).

An individual with their alienable possessions:

e. John replaced his computer / his trainer / his glasses.

Container with content:

d. Joe replaced the water in the vase.

Temporary configuration:

f. Mary replaced the bag on the desk.

The examples show that the notion of a structured whole is independent of single objecthood.²

A whole need not be an object; but may be a configuration, a plurality of objects standing in particular relations to each other.

There is a constraint that the whole (single object or configuration) be a structured whole. Thus, the following a-examples are excluded:

(5) a. ??? Sue replaced the flower she had just picked.

b. Sue replaced the orchid in the flower arrangement.

(6) a.??? John replaced the person speaking to his wife.

b. John replaced the person chairing the session.

(7) a. ??? John replaced the person that was spontaneously addressing the crowd.

b. John replaced the first speaker at the conference.

Replacement can also target roles not contained within specific particular wholes:

² See Moltmann (2025) for a discussion of this point.

- (8) a. John replaced the housekeeper.
 b. John replaced his umbrella.

There are constraints, though, on what may constitute such roles. In general, it appears roles need to be connected to a fixed independent entity, what Langacker's (1993) calls a 'reference point'. In fact, replacement bears a close semantic connection to 'reference point constructions', in Langacker's sense, namely the *have*-relation and possessive construction:

- (9) a'. John's umbrella
 b.' the umbrella John had

That is, the independent entity 'has' the replace / replacer.

Replacement as conveyed by *replace* targets a part of a structured entity that occupies particular slot *within* that entity. *Replace* cannot target the temporal or spatial location of the entity itself:

- (10) a. ???John replaced the location of the company.
 b. ??? Mary replaced the time of the ceremony.

The reason is a metaphysical one. Entities have a location in time and space, but that location itself does not occupy a slot within the entity.

Let us call the structured whole involved in the interpretation of replace the target of *replace* (of replacement) and the entity or portion that is subject to replacement and occupies a slot in the structured whole the entity *replace* (replacement) applies to.

The verb *exchange* is another verb of replacement. Sometimes it is interchangeable with *replace*:

- (11) a. John replaced the screw.
 b. John exchanged the screw.

Sometimes it is not:

- (12) a. John exchanged the speakers

- b. John replaced the speakers
- (13) a. John and Mary exchanged addresses
 - b. John has Mary's address.
 - c. Mary has John's address.

The difference resides in that *replace* targets entities for a single slot, whereas *exchange* targets entities a and b, occupying distinct slots.

Another consequence is that *exchange* cannot apply to something like *the missing screw*, a variable object that fails to have an actual realization:

- (14) a. John replaced the missing screw.
- b. John exchanged the missing screws.

How is a slot to be understood as a component of a structured whole? The philosophical literature offers two potential candidates for how to understand the notion of a slot formally, as something that can be occupied by different entities at different times (and perhaps different circumstances more generally).

The first is Fine's (1999) notion of variable embodiment. A variable embodiment is an entity that is associated with a function mapping a time (or circumstance) to a concrete manifestation of that entity at the time of circumstance. The notion of a variable embodiment was precisely introduced in order to account for the replacement of parts or matter. 'The water in the container' and 'the president of the US' thus are considered variable embodiments allowing for different portions of water and different individuals as manifestations at different times. Replacement targets slots whose fillers are not themselves essential to the whole, which in Fine's (1999) terms means that replacement targets rigid embodiments consisting of variable embodiments. A common critique of the notion of a variable embodiment, though, is that it imposes no constraints whatsoever on the manifestation function or the structure of the manifestations. Variable embodiments do not come for free, rather they arise on the basis of a structure being individuating of an object.

A second notion proposed in the literature is the notion of a slot in slot mereology (Bennett 2013). Slot mereology originally developed for abstract wholes such as situations and structural universals, which permit the same entity occurring as a part multiple times. On that view, an entity may be composed of slots, so that the part relation P is relativized to a slot, which permits

the same entity to occur as part of an entity several times over. It is straightforward to extend the notion of a slot in slot mereology so that an entity is not just part of another relative to a slot, but also relative to a time, thus allowing for replacement with respect to the same slot over time. The notion of a slot in slot mereology is also subject to critique: there are no constraints imposed on slots. Slots are viewed as entities on their own, without connection to the structure of the whole being imposed. But then can slots as entities in themselves just compose an object?

To address that critique, we should ask the question, what is it that underlies a slot or constitutes a slot in a structured whole? For that we should address the question: what is a structured whole? For Aristotle, structured wholes are composed of matter and form. This leaves open the status of form itself. Let us assume, with Johnston (2007) that form itself consists in various essential relations or properties that obtain among or of parts: they constitute the ‘glue’ that keeps the whole together. Then slots arise from the various positions associated with the relations or properties constitutive of a structured whole.

Positing positions of relations is the view of positionalism discussed in the metaphysics of relations (Fine 2000). Fine introduced positionalism as a possible position in the metaphysics of relations, after having established relations as being neutral, not imposing any direction among arguments (see also Williamson 1985). Thus, a binary relation and its converse can only be completed as the very same state of affairs. The cat being on the cat and the mat being under the cat are the very same state of affairs. The idea is that relations are associated with a number of positions which are distinct from each other, but not ordered. Relations then are completed by entities relative to particular positions. Positionalism has a correlate in linguistics, in that relational predicates are associated with theta roles (without order among them), to be assigned to occurrences of expressions in the sentence.

Fine himself did not adopt positionalism. One reason being symmetric relations (which I will set aside); the other being the apparently obscure ontology of positions, which positionalism posits as primitives.³

There are several points to be made regarding the skepticism about the ontology of positions. First, positions are on a par ontologically with roles such as ‘the president of the US’, ‘our home’, etc., and we accept roles in our ontology explicitly.⁴ We have nouns for particular roles. The noun *home* is specific to a role; nominals like *president of the US* can be used for roles.

³ There are proposals for dealing with symmetric relations within positionalism, such as reanalysing symmetric relations as plural properties (Dixon 2018).

⁴ Roles just are generally associated with additional conditions defining their social importance.

Positions of relations are not in principle distinct from roles, rather they belong the same general category. In fact, we have seen that replacement may target roles without a structured whole (but just a reference point). Second, positions appear to be needed in order to account for the possibility of conjoining relations with particular argument positions, in examples such as:

- (15) a. This the friend of Bill and teacher of Mary.
 b. John is happy and healthy.

In linguistics the operation of conjoining relations with respect to particular positions is known as ‘theta-identification’ (Higginbotham 1985). Positions are needed also in linguistics, of course, in order to ensure the alignment with occurrences of noun phrases in the sentence with the lexical meaning of the predicates (theta-roles). In addition, in linguistics, generally a particular position of a predicate is distinguished as the ‘external’ argument position: the external argument position is aligned with the subject and specifier position; where internal positions align with direct and indirect objects. Positions, finally, are reflected overtly in natural language, with the (light) noun *place* as in *replace*, *in your place*, and *in-stead*, in *English* (and similar observations can be made across languages).

Slots can then be taken to consist in the ‘identification’ of positions of different relations or properties constitutive of the structure of structured wholes. Instead of starting with the notion of a slot itself, the point of departure thus will be with is constitutive of a slot, namely structure. A structure consists of various relations among entities making up the parts of the whole at a time. Relations in turn do not just apply to entities, but to entities relative to positions.

For the semantics of replacement, I will therefore make use of positions (using the term for a larger category that includes roles. The verb *relace* describes acts of replacement, which may involve an agent, a replacee and a replacer or just a replacer and replacee:

- (16) a. John replaced the chairman.
 b. John replaced Bill as chairmen.

The semantics of *replace* (in the first sense) then involves the following condition:

- (17) Replacement applied to structured entities

For a structured whole w with a slot l occupied by an object a , for an event e and agent a , if $\text{REPLACE}(e, d, a, b, l, w)$, then a occupies l at the outset of e and b occupies l as a result of e .

2. *Instead*-phrases

Instead-phrases like verbs of replacement involve replacement as an operation on structured wholes. But *instead*-phrases do not apply to structured wholes that are ordinary objects, but rather structured wholes that are semantic objects, objects generated by semantic composition or pragmatic processes, namely, structured propositions (or states of affairs) and background situations. Four types of *instead*-phrases can be distinguished

2.1. *Instead*-phrases of type 1

Instead-phrases of type 1 take various complements (NPs, APs, PPs, gerunds) and relate to particular constituents in the sentences by *association with focus* (Rooth 1995):

- (18) a. Instead of *Mary*, Bill met *Sue*.
 b. Instead of a *man* the firm hired a *woman*.
 c. Instead of *running*, John *walked* to the house.
 d. Instead of *over* the house, the bird flew *into* the house.
 e. Instead of giving *John a book*, Mary gave *Bill a pen*.
 f. Instead of *the children*, Mary met with *the parents*.

Syntactically, type 1 *instead*-phrases seem to engage in various constructions: subordination (PPs), as in (19a), comparatives as in (19b), and coordination, as in (19c):

- (19) a. In place of John, they hired Mary.
 b. Rather than walking to the house, John ran toward it.
 c. The bird flew into rather than /instead of over the house.

Instead-phrase apply to structured wholes, structured meanings of sentences that come with slots. The slots in such structured proposition filled by the semantic values of the focused constituents.

There are general differences between the replacement conveyed by *replace* and conveyed by *instead*. First, *instead* may involve higher-order semantic values and plural semantic values as replacers. The replacer / replacee may be

- an object (18a)
- a higher-order semantic value, e.g., a quantifier (18b) or a relation (18c, d)
- an ordered plurality of objects, relating to distinct positions in the structured proposition at once (18e)
- an unordered plurality (18f).⁵

Second, *instead*-phrases, unlike *replace*, may relate to spatial and temporal locations:

(20) a. Instead of *Monday*, they arrived *Tuesday*.

b. Instead of in the *garden*, they celebrated in the *house*.

The reason is that temporal and spatial locations may occupy places in a structured proposition; they are not locations of structured propositions, unlike in the case of ordinary objects.

For the purpose of this paper, I will make use of standard alternative semantics, as in Rooth (1993) for the semantics of *instead*-phrases. This permits a simply way of formulating the replacement conveyed by *instead*: *instead* involves mapping the focus-semantic value of the sentence applied to the replacee to the focus-semantic value of the sentence applied to the replacer:

(21) The semantics of type 1 *instead*-phrases (*instead*-phrases with association with focus)

$[Instead\ of\ X_F [\dots Y_F \dots]_S]^w = 1$ iff for the focus-semantic value of p of S , $p^w([Y]) = 1$,
provided that $p^w([X]) = 0$ and for any world w' in the common ground, $p^{w'}([X]) = 1$.

⁵ The ability of *instead* conveying the replacement of higher-order objects has to do with the linguistic fact that *instead* is not restricted to nominal complements (like *replace*), but may act as a coordinator (*over instead of under*) or as a comparative, like the synonymous *rather than*, which applies across categories (*blue rather than yellow, a man rather than a woman* etc.).

Instead-phrases of type 2 involve an expectation or consideration regarding the proposition in which the semantic value of the complement of *instead* fills in the slot in question. In (20), such a proposition is taken to be part of the common ground, on a suitable weak notion of common ground.

2.2. *Instead*-phrases of type 2

Instead-phrases of type 2 take a clausal complement, in English a gerund, as in (22a-c) or, for example in German, a *that*-clause, as in (22d):

- (22) a. Instead of raining, it snowed.
 b. Instead people dancing in the street, there was silence everywhere.
 c. Instead of John praising himself, Joe criticized Mary.
 d. Statt dass es regnete, schneite es.
 instead that it snowed, it rained

They do not associate with focus and relate to particular positions in the sentence. Semantically, they relate to a slot-providing situation that is entirely given by the context (background situation). The background situation was expected to be filled by a subsituation or an event (as described by the complement of *instead*) and is filled by an alternative event instead. Type 2 *instead*-sentences seem to be restricted to event descriptions:

- (23) a. ??? Instead of knowing / speaking French, John knows / speaks Spanish.
 b. Instead of speaking French, John spoke Spanish during the meeting.

Background situations as structured wholes contain a slot individuated by spatio-temporal location and concurrence with other events or states.

What distinguishes structured propositions and background situations from ordinary structured wholes? The former are generated by semantic or pragmatic processes; the latter are part of our ordinary ontology. More precisely, structured propositions result from semantic composition (of focus-semantic values); background situations result of pragmatic processes (serving pragmatic enrichment).

For the formal semantics of type 2 *instead*-phrases, I will modify alternative semantics so as to apply to background situations, that is situations that contain a slot that can be filled by another subsituation. The result is then (24), where $\text{pos}(S)$ is the set of situations described by the sentence S , or the set of exact truthmakers of S in the sense of Fine (2017):

(24) The semantics of instead-phrases of type 2 (*instead*-phrases without association with focus)

For a contextually given situation c with a slot l for a subsituation s ($c(l/s)$),
 $[\textit{instead of } S', S] = 1$ iff there is a situation s' , $s' < w$, $s' \in \text{pos}(S)$, $\lambda s[c(l/s)](s') = 1$,
 provided $s \notin \text{pos}(S)$ and there is a possible situation s'' , $s'' \in \text{pos}(S')$,
 and $\lambda s[c(l/s)](s'') = 1$.

On this account, type 2 *instead*-phrases are evaluated relative to a background situation containing a slot filled by a situation described by the complement of *instead*. Of course, such a background situation may be viewed as part of the common ground, allowing for an assimilation of type 1 and type 2 *instead*-phrases.

2.3. The anaphoric adverb *instead*

Instead can also be used on its own, without complement, as an anaphoric adverb.

Instead without complement when associating with focus acts as an adverbial that relates anaphorically to the replacee in a preceding negative sentence:

- (25) a. Bill did not meet *Mary*. *Instead*, he met *Sue*.
 b. The firm should have hired a *woman*. *Instead*, it hired a *man*.
 c. John did not *run* to the use. *Instead*, he *walked* to the house
 d. The bird did not fly *over* the house. *Instead*, it flew *into* the house.
 e. Mary did not give *John a book*. *Instead*, she gave *Bill a pen*.

The adverb *instead* in a sentence S requires the preceding sentence S' to entail or implicate the negation of what S describes applied to the replacee rather than the replacer.

Instead without complement when not associating with focus only requires the preceding sentence to entail or implicate the negation of S :

- (26) a. It should have rained. Instead, it snowed.
 b. People did not dance in the street. Instead, there was silence everywhere.
 c. John did not praise himself. Instead, Joe criticized Mary.
 d. ? John does not know French. Instead, he knows Spanish.

Instead on that use is evaluated relative to a contextually given proposition. By taking anaphoric *instead* to refer to such a proposition, it can then be given the same semantics as *instead*-phrases relating to a background situation:

(27) The semantics of anaphoric *instead*

- a. For the proposition of the form $\lambda w[p^w(d) = 1]$ that *instead* refers to,
 $[Instead \dots Y_F \dots]_S^w = 1$ iff for the focus-semantic value of p of S , $p^w([Y]) = 1$,
 provided that $p^w(d) = 0$ and for any world w' in the common ground, $p^{w'}(d) = 1$.
 b. For a contextually given situation c with a slot l for a subsituation and a proposition p
 (set of situations) that *instead* refers to,
 $[Instead S] = 1$ iff there is a situation s' , $s' < w$, $s' \in \text{pos}(S)$, $\lambda s[c(l/s)](s') = 1$, provided
 $S' \notin \text{pos}(S)$ and there is a possible situation s'' , $s'' \in p$ and $\lambda s[c(l/s)](s'') = 1$

2.4. Replacement with descriptions

Instead-phrases in descriptions are of particular interest and require a different semantics:

- (28) a. the student that uses a typewriter instead of a computer
 b. the house that has a balcony instead of a garden
 c. the spoon that is made of wood instead of silver
 d. the bird that flew into instead of over the house

In such cases, the scope of the *instead*-phrase is the description itself (an NP), which means that the object described will be the entity whose description involves the replacement in question. *Instead*-phrases in descriptions pose a considerable challenge to standard semantics,

that is, a semantics on which descriptions just denote objects. The replacements described in (28) can hardly operate on concrete objects. First, there need not be a concrete object of which the description with the replacee holds. Second, the replacee and the replacer may be higher-order objects, which by nature could not be parts of concrete objects .

The structured wholes on which *instead*-phrases operate here are expected types, as in (28), kinds as in (29a) as well as previous or future types of stages of objects as in (29b, c):

- (29) a. the swans that are black instead of white
 b. the house that now has two doors instead of one
 c. the house that will be blue rather than yellow

For the formal semantics, one may again make use of alternative semantics. The focus-semantic value of the nominal will then be a function from semantic values to descriptions of types of objects. The replacement will consist in mapping a background type (with the replacee) onto a type (with replacer) that also serves as a description of particular individuals.

The semantics of *instead*-phrases in descriptions involves two abstract wholes containing the same slot(s). Such abstract wholes permit replacement of higher-order objects as components and can, but need not, be manifested by concrete objects. The description itself will be evaluated as denoting the manifestation of the abstract whole with the replacer.

3. *Place*-NPs

I will only finally briefly mention *place*-phrases as replacement expressions as below:

- (30) a. In your place, I would have declined the offer.
 b. ?? In your place, the company would have hired Bill.

Place-phrases come with a counterfactual conditional force and generally relate to the subject of the sentence. *Place*-NPs involve a slot in a contextually given situation, like type 2 *instead*-phrases. What is special about *place*-phrases is the connection between antecedent and consequent, which is established on the basis of empathy or simulation. This is reflected in the equivalence of *place*-phrases as in (30a) to the antecedent of an empathetic conditional as below:

(31) If I were you, I would have declined the offer.

Here ‘you’ refers to the addressee as someone occupying the slot in the contextually given situation.

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